UF1. Introduction to HTML5

- Review of key concepts
- Introduction to HTML Language
- HTML5 features
- Structure of a HTML5 document
- HTML5 Semantic containers



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Warm-up…

Pairs. Discuss and contrast these pairs of concepts.

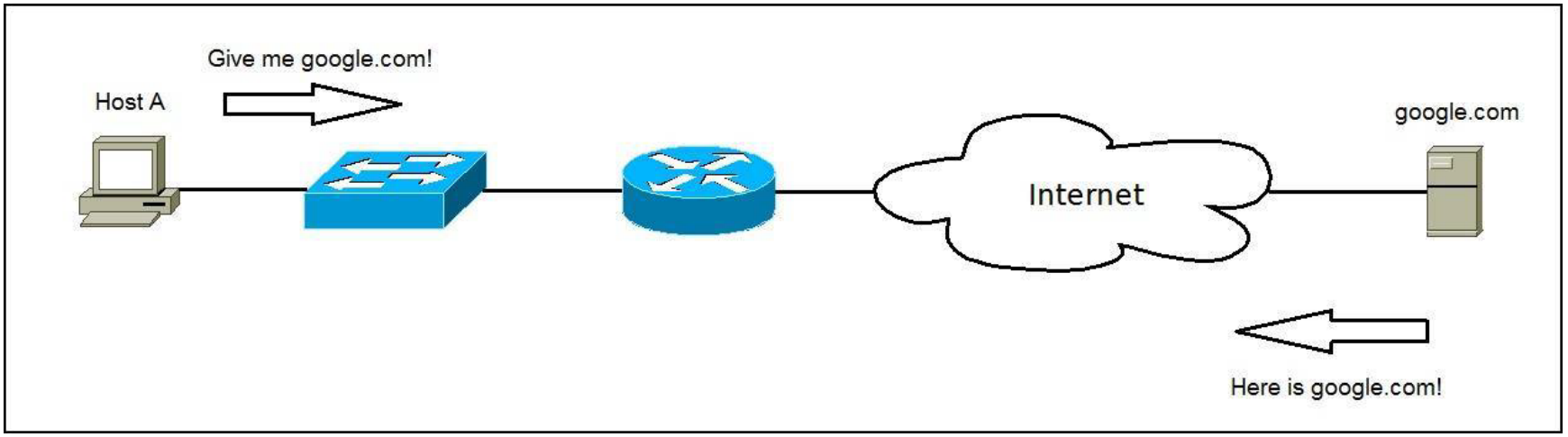
Find differences.

1. **Internet** **vs** **World Wide Web**
2. **HTML** **vs** **HTTP**
3. **Web page** **vs** **Web site** **vs** **Web app**
4. **Static page** **vs** **Dynamic page**
5. **HTML code** **vs** **Script code**
6. **Client script** **vs** **Server Script**

1. Review of concepts: WWW Service

- The *World Wide Web* *(also* WWW or W3) is a **service** that allows remote information access, based in:
  - The protocols for data transferring and addressing on the Internet: TCP/IP

    Note: ”Internet” ≠ ”WWW”
  - Language HTML: HYPERTEXT documents, that allows access and browsing resources linked (HYPERLINKS)
  - A Client / Server model, using two other protocols:
    - Documents are stored in Servers, and they are requested using their URL (*Unified Resource Locator*)
    - The protocol HTTP (*Hypertext Transfer Protocol*) is used for transmission of documents required by the Client (web *browser*)



Layer #	Layer Name	Protocol	Protocol Data Unit
5	Application	HTTP, SMTP, etc…	Messages
4	Transport	TCP/UDP	Segments/ Datagrams
3	Network or Internet	IP	Packets
2	Data Link	Ethernet, Wi-Fi	Frames
1	Physical	10 Base T, 802.11	Bits

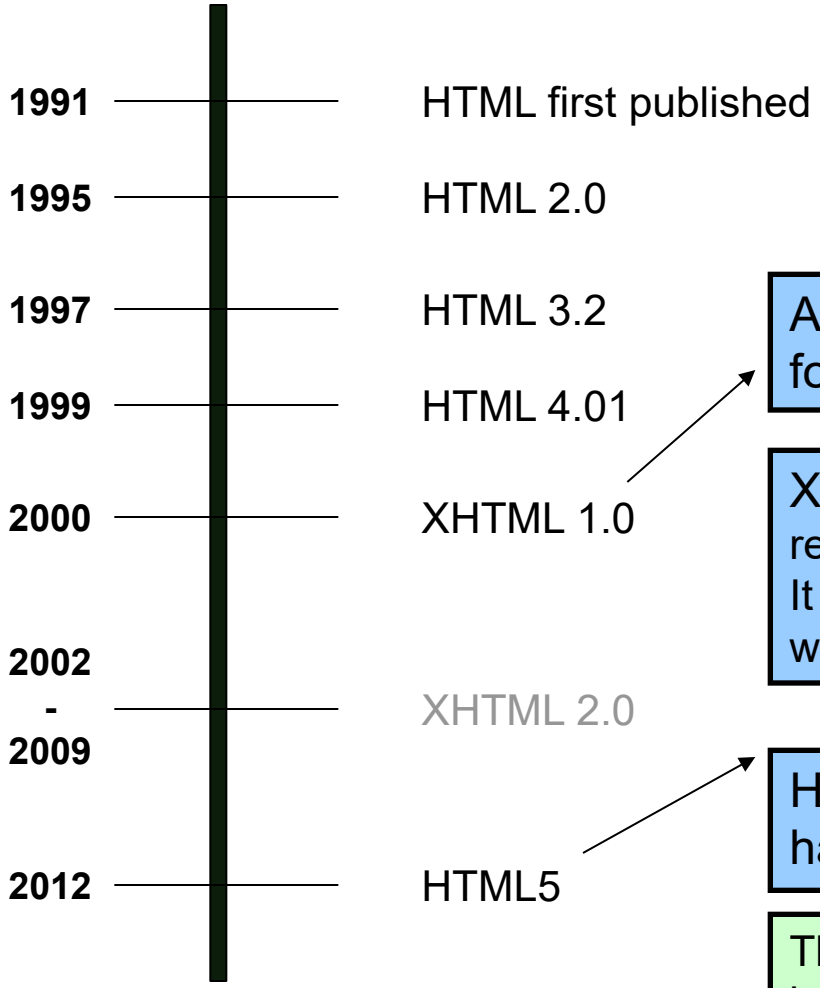
https = http + SSL (secure socket layer)



2. Introduction to HTML Language

- HTML: Hypertext Markup Language
- The standard markup language for creating web pages and web application
- First description of HTML:
  - Published by Tim Berners-Lee in 1991
  - Scientist at CERN, invented the WWW (url, http, html) for automatic information-sharing between scientists in universities and institutes around the world
  - Only 22 elements, and included the basic structure of an HTML doc.
- Based on SGML (Standard Generalized Markup Language), which was also based on IBM - GML.
- HTML Semantic and Standards is defined and maintained by the WWW Consortium (http://www.w3.org) and later by WHATWG

2. Introduction to HTML Language



After HTML 4.01 was released, W3C was focused stricter standards as XHTML.

XHTML 2.0 had stricter standards than 1.0, rejecting web pages that did not comply.
It was not well accepted by many companies and was abandoned completely in 2009.

HTML5 is much more tolerant and can handle markup from all the prior versions.

Though HTML5 was published officially in 2012, it has been in development since 2004.

2. Introduction to HTML Language

- Documents written in **HTML** must have extension: **.html** o **.htm**
- Content must be written in text-plain (ascii)
- HTML is a MARK-UP language:
  - ELEMENTS, to define the contents of documents and sections. Examples: **<body>...</body>**
    ** Un texto **
  - Each element (paragraph, table, image, etc.) is declared with:
    - "Opening TAG" **<p**
    - some (or none) ATTRIBUTES **id="par1">**
    - the CONTENT **Un texto**
    - "Closing TAG" ** **
  - The ATTRIBUTES specified inside opening label
    …
- TAGS can be written in **Lowercase** (recommended) and Uppercase
- Two kinds of elements : <title>...</title>
  "empty elements"

3. HTML5 overview

- W3C raised the first version of HTML5 in 2012
  The World Wide Web Consortium (W3C) is a consortium of companies and organizations, which maintain full-time staff for the purpose of working together in the development of standards for the WWW.
- Newest version: HTML 5.2, from December 2019, “work in progress”

  https://html.spec.whatwg.org/

  Note: w3c and WHATWG converged in 2019
- It’s a “recommendation”, supported by the makers of web browsers. So it’s nearly like the “standard”.
- Incorporates all features from earlier versions of HTML, including the stricter XHTML.
- HTML5 is a “work in progress”. No browsers have full support, but they are in the way + check the support: https://caniuse.com

3. HTML5 overview

- Goals:
  - Support all existing web pages
  - Reduce the need for external plugins and 3rd part scripts
    HTML5 has integrate javascript as the main client script language
  - Improve the semantic definition of page elements
  - Rendering of web content universal and independent of the device being used (media responsive)
- New features:
  - New tags: <meter>, <dialog>, <template>
  - Built-in audio and video support (without plugins) : <audio> <video>
  - Improved form controls and attributes: <input type="**email**">
  - Element **Canvas** (a way to draw directly on a web page) <canvas>
  - Support for CSS3 (the newer and more powerful version of CSS)
  - New API's: Drag and Drop functionality, Geolocation, Web Sockets…
  - Advanced features, such as data storage and offline applications

3. HTML5 overview

- Everything between a “opening tag” and a “closing tag” is called **element**:

  *The **elements** represents structures to organize the document content.*

  *There are elements that provide special functionality when browsing: <a href…> *

- There are “**empty elements**”, but we have to close them:

  <meta charset=“utf-8” /> **
** *(note: we have just a “opening&closing” tag!)*

- Elements could have **attributes**.

  *An **attribute** is a pair **name=“value”**, inside de opening tag.*

  The attributes can give properties and behaviour to the elements.

  - Basic Attributes: **id=**“nom” **class=**“lateral” **style=**“display:none” **name=**“imagen”
  - Event Attributes: **onclick=** “validate()” onchange, onfocus, onload …
  - Specific Attributes of a tag: <a **href=** … > <img **src=** … > <input **disabled** >

+info: http://www.w3schools.com/html/html5_intro.asp http://es.wikipedia.org/wiki/HTML5

4. Basic structure of a HTML5 document

- An HTML document (or webpage) has three basic elements:

```
<!DOCTYPE html>
<html lang="en">
 <head>
  ... header of html document ...
 </head>
 <body>
  ... body of  html document (content of the webpage)..
 </body>
</html>
```

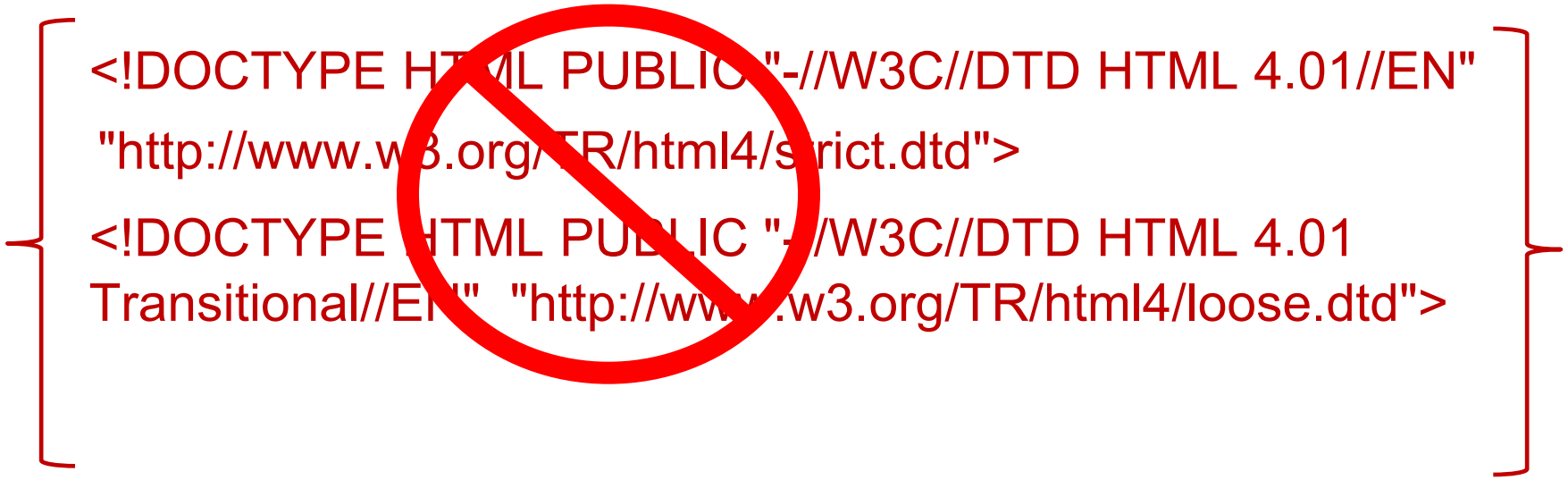
lang attribute declares the language of the content.
It is not strictly required, but it should be specified. It can help search engines and screen readers.

https://www.w3.org/International/questions/qa-html-language-declarations.es

https://www.w3schools.com/Tags/ref_language_codes.asp

4.1 DOCTYPE directive

The DOCTYPE tells the browser which type and version of document to expect.



- In **HTML5**, the declaration is very easy:

<!DOCTYPE html>



4.2 Header of the document **<head>**

- This element is the header of the document HTML

  <head> ... </head>
- The “head” is the NON-VISIBLE part of the document
- Inside this “head” we can declare different things:
- <title>...</title> to set the title of the page
- <base … /> URL base (when using relative addesses)
  - <base href="http//ants.dif.um.es/asignaturas/tasi“ />
- <link … /> to link a stylesheet document, for example…
  - <link rel="stylesheet" type="text/css" href="css/global.css" />

4.2 Header of the document **<head>**

- **<meta>** provides information about the document (useful for seach engines)
  - <meta name="author" content = "Eduardo Martinez“ />
  - <meta name="keywords" content="music pop fiesta spain" />
  - ~~<meta http-equiv="content-type" content="text/html; charset=utf-8" />~~

  In HTML5.0, the tag to specify the **character set** has been simplified:



<meta charset=“utf-8" />

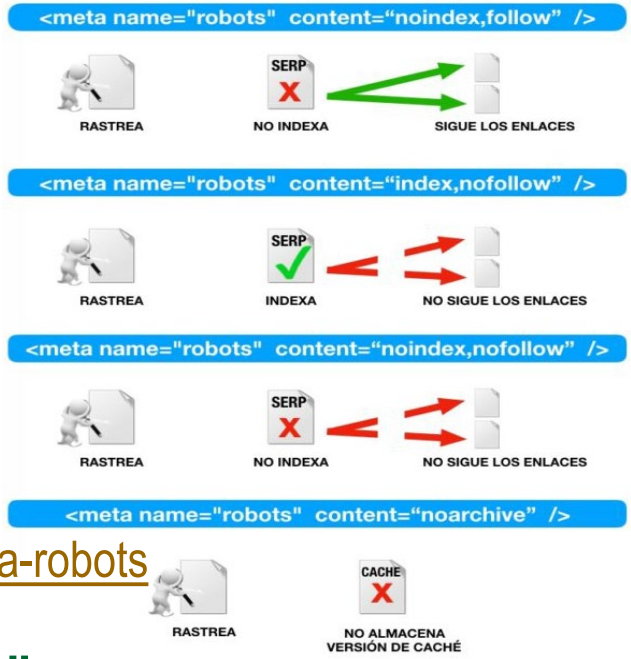
- Other uses of **<meta>**

  Force navigator to reload the page each 5 seconds:
  - <meta http-equiv="refresh" content="5" />

  Force navigator to load the page specified, after 10s:
  - <meta http-equiv="refresh" content="10; URL=http://www.google.es" />

  Establish a expiration date. After this, the navigator cannot get the page from the cache:
  - <meta http-equiv="expires" content="Fri, 05 Jan 2008 13:00:00 GMT" />

4.2.1 Nuevos usos de <meta>



- **<meta name="robots" content=" ">**
  - Indica a los robots de búsqueda si deben indexar la info de la página
  - Por defecto si no la ponemos, los robots indexan, así como siguen enlaces
  - Podemos customizarlo:

https://www.humanlevel.com/diccionario-marketing-online/meta-robots

- **<meta name="theme-color" content="#ff6600">**
  - Fija el color de tema de navegadores Chrome, firefox y Opera
    https://developer.mozilla.org/en-US/docs/Web/HTML/Element/meta/name/theme-color

- **Metaetiquetas Open Graph: metatags sociales** http://ogp.me
  - Ayuda a que noticias y blogs se posicionen mejor para su tópico específico
  - Además favorece su difusión dentro de las redes sociales

```
<meta property="og:type" content="article" />
<meta property="og:title" content="TÍTULO DE MI POST Ó PÁGINA" />
<meta property="og:description" content="DESCRIPCIÓN DEL CONTENIDO" />
<meta property="og:image" content="LINK A LA IMAGEN DESTACADA" />
<meta property="og:url" content="ENLACE PERMANENTE" />
```

4.3. Body of the document **<body>**

- This element is the body of the document HTML

  <body> ... </body>

- The “body” is the VISIBLE part of the document html.
- Inside the <body>, we will include the elements that give content and structure of the webpage.
- In older versions of HTML, body could have attributes:
  ~~<body text="#FFFF66" link="#0000FF" vlink="#FF0000">~~ 

  Note: In HTML5, we could not specify any presentation attribute, apart from the standard, used to link css properties (p.e. style=“color: grey”*)*

- One of the main goals of HTML5 is
  “Separate SEMANTIC from PRESENTATION (style+layout)”

  With HTML5 we structure info semantically, with CSS3 we will give style

- So, inside the body element, we have to include tags as:
  <header>, <nav>, <section>, <article>, <aside>, <footer> …

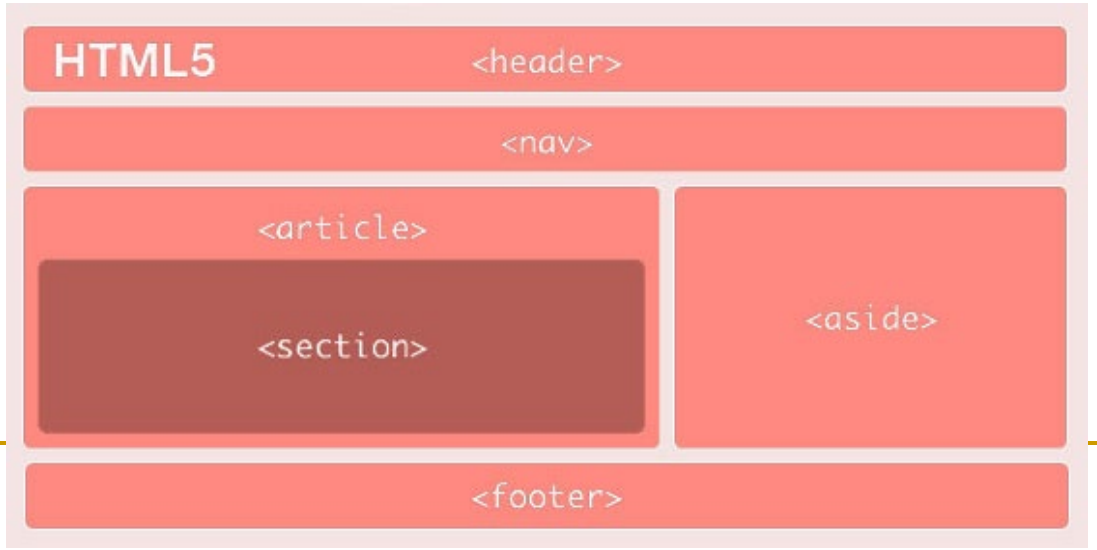
5. Semantic containers inside the **body**

HTML 4.01: just div's with id attribute:

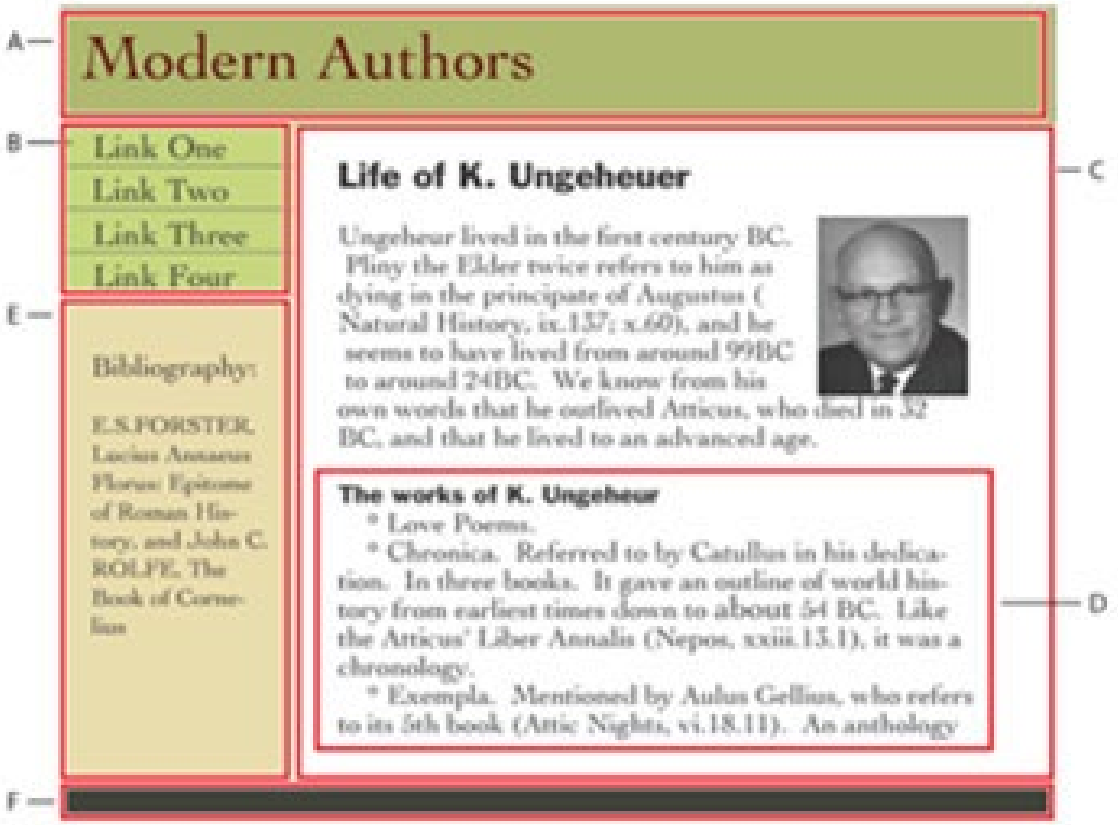


HTML5: diferente container tags. It makes more sense… SEMANTIC web!





5. Semantic containers inside the **body**



A. header. B. nav. C. section. D. article. E. aside. F. footer.

5. Semantic containers inside the **body**

We have many new HTML5 tags, to declare different containers.

We must use them to give the best structure of the content:



- **<header></header>**: To place content of the header (visible header)
- **<main></main>:** Main container of the document, or main functionality
- **<nav></nav>**: Navigation section of the page (where we'll find hyperlinks).
- **<section></section>**: To group content related, o to mark different sections of the document.
- **<article></article>** Content that could be published and shared (blog entries, articles)
- **<aside></aside>**: To mark an área that includes content not very important or off the topic of the page (banners, widgets, weather, calendars….)
- **<footer></footer>**: the footer of the page (author info, credits, copyright).

+ info: https://11bits.es/etiquetas-estructurales-en-html5/

Note: *** *** *and* ****** *containers, could also be used.*
But always is better to use specific HTML5 container tags.

5. Semantic containers inside the **body**

```
<!doctype html>
<html>
   <head>
      <title>T&iacute;tulo de P&aacute;gina</title>
   </head>
   <body>
      <header>
         <h1>T&iacute;tulo de P&aacute;gina</h1>
      </header>
      <nav>
         <!-- Navegación -->
      </nav>
      <section id="intro">
         <!-- Introducción -->
      </section>
      <section id="intro">
         <!-- Contenido principal -->
      </section>
      <aside>
         <!-- Barra Lateral -->
      </aside>
      <footer>
         <!-- Pie de Página -->
      </footer>
   </body>
</html>
```

Extra: Writing contents

To write content inside the different containers,
we have the same tags tan in HTML 4.01

- **<h1>..</h1> <h2>..</h2> …. <h6>..</h6>**

  <h1>..</h1> : tags for title headers (6 levels)
- ** ... **: paragraph
- **
**: line break
- **<hr />**: horizontal row
- **<!-- comment -->**: comment (not visible)

Extra: Writing contents

- Special characters:

Caracter	Códigos	Caracter	Códigos	Caracter	Códigos
á	á	Ú	Ú	<	<
Á	Á	Ü	ü	>	>
é	é	Ü	Ü	&	&
É	É	Ñ	ñ	"	"
í	í	Ñ	Ñ	º	º

- ** ** Space character ("Non Breaking Space")
  - To include more than one space between two words
  - To avoid two words to be splited between two lines of the paragraph